



Green separation of rare earth elements by valence-selective crystallization of MOFs†

Pengfei Yang,  Qixin Zhuang,  Yongsheng Li  and Jinlou Gu  *Cite this: *Chem. Commun.*, 2019, 55, 14902Received 7th October 2019,
Accepted 18th November 2019

DOI: 10.1039/c9cc07849e

rsc.li/chemcomm

Herein, a novel valence-selective crystallization (VSC) strategy is developed for the precise separation of Ce element from rare earth (RE) mixtures by selectively crystallizing Ce(IV) into pre-designed metal–organic frameworks (MOFs). The designed VSC strategy features fast separation speed, mild separation conditions, pollution-free operation, high separation efficiency, a wide selection range of ligands and extensibility to other elements.

The unique optical and magnetic properties of rare-earth (RE) elements, including lanthanide (Ln) elements, Y, and Sc, are widely exploited in various advanced fields of science and technology.¹ Among all RE elements, Ce is particular not only because of its wide applications in catalysts,² luminescent materials,³ UV absorbents and UV filters,⁴ and in many other fields,⁵ but also because of its stable 4-valence form while other RE elements generally exist in a 3-valence state. Therefore, the precise separation of Ce from a RE mixture is of great significance. However, it is very difficult to separate solely Ce from other RE elements due to their similar chemical and physical characteristics. Although various methods, such as solvent extraction,⁶ ion chromatography,⁷ and solid adsorption,⁸ have been developed for RE separation, a technique with the advantages of convenience, short separation time, low-cost, high separation efficiency and environment benignity has not yet been established. Thus, it is highly desirable to develop a green and reliable strategy for the accurate separation of RE mixtures.

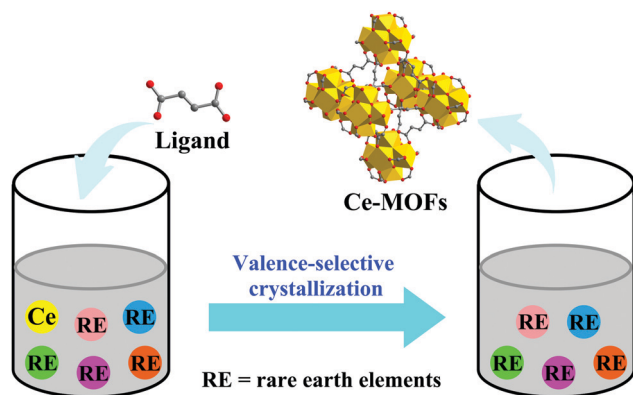
Recently, as a new class of porous crystalline materials, metal–organic frameworks (MOFs) constructed from metallic nodes and organic linkers through coordination bonds, have attracted wide attention due to their fascinating properties, such as designable structure, adjustable pore size and abundant structure.⁹ Interestingly, the specific topological structure of the MOFs could be

regulated by numerous factors, such as ligand configuration, metal ion size and metal ion valence states, which usually requires precise matching between metal ions and organic linkers. Based on the mutually selective characteristic between organic linkers and metal ions, some interesting strategies have been developed for the separation of RE ions or isomeric organic ligands by selective crystallization of MOFs.¹⁰ For example, we successfully realized the selective separation of isomeric dicarboxylic acids by selectively crystallizing the target isomer into a pre-designed MOF.^{10a} Bu's group developed a size-selective crystallization strategy to separate the RE element,^{10d} Zheng's group successfully extracted Nd from other RE elements by the size-selective crystallization strategy.^{10b} Although these studies provide new ideas for RE separation, the development of a precise and green separation approach for RE elements is still in its infancy.

Based on the fact that the separation and purification of Ce mainly depend on the organic solvent extraction method with high cost and pollution,¹¹ combined with the above conception of selective crystallization of MOFs, we developed a simple, effective and pollution-free valence-selective crystallization (VSC) strategy for the precise separation of Ce by selectively crystallizing Ce(IV) into pre-designed Ce(IV)-MOFs from RE mixed solutions (Scheme 1). To avoid toxic reagents involved in the synthesis of pre-designed Ce(IV)-MOFs, an aqueous synthesis of Ce-MOF-801 constructed with Ce(IV) and non-toxic fumaric acid (FUM) is first explored in detail. Subsequently, based on the green synthesis of Ce-MOF-801, Ce⁴⁺ and other RE³⁺ ions are chosen as the model targets to verify our hypothesis of the VSC strategy. As expected, Ce(IV) can be selectively crystallized to Ce-MOF-801 using a FUM ligand (Scheme 1). Thanks to the strong crystallization ability between tetravalent Ce and the FUM ligand, the VSC process can be completed in 10 min at room temperature, which saves a lot of time and energy costs. More importantly, only trace RE elements (less than 0.03% based on Ce) are detected in Ce-MOF-801 obtained from the RE/Ce separation system, indicating the effectiveness and feasibility of the VSC strategy. The developed VSC strategy remains effective upon enlarging the designed separation system to 100-fold, suggesting its great potential for industrial applications. It is worth

Key Laboratory for Ultrafine Materials of Ministry of Education, School of Materials Science and Engineering, East China University of Science and Technology, Shanghai 200237, China. E-mail: jinlougu@ecust.edu.cn; Fax: +86-21-64250740; Tel: +86-21-64252599

† Electronic supplementary information (ESI) available: Experimental details, XRD patterns, SEM images, N₂ sorption data, Tables S1 and S2, molar fraction data. See DOI: 10.1039/c9cc07849e



Scheme 1 Schematic illustration for the separation of Ce^{4+} from other RE^{3+} ions by a valence-selective crystallization strategy.

mentioning that the VSC strategy features a wide selection range of ligands. Besides FUM, sodium 1,4-benzenedicarboxylate (Na_2BDC), 1,3,5-benzenetricarboxylic acid (BTC) and succinic acid (SUC) can also be successfully applied to the RE separation system, and their corresponding Ce-MOFs (Ce-Uio-66, Ce-MOF-808 and Ce-Uio-66-SUC) were also obtained for the first time from aqueous solution at room temperature. In addition, the VSC strategy is universal to other separation systems such that it could be extended to the precise separation of Fe from the mixed solutions of Fe^{3+} , Ni^{2+} , Co^{2+} and Cu^{2+} ions. Therefore, the fast, mild, efficient, and environmentally friendly VSC strategy provides a new perspective for the accurate separation of Ce or other elements.

To realize a green and low-cost Ce separation by the designed VSC strategy, the pre-designed Ce-MOF-801 with a 12-connected three-dimensional framework (Fig. 1A) was constructed by choosing water as a solvent and inexpensive FUM as a ligand.¹² Furthermore, the correlation between the most significant parameters such as crystallization temperature, the amount of acetic acid (HAc) modulator as well as the crystallization time was investigated in detail.

Temperature is an important parameter in the crystallization process of MOFs,¹³ which might affect the efficiency of Ce^{4+} ions

entering Ce-MOFs. All Ce-MOF-801 samples prepared in different temperatures display XRD patterns identical to the simulated Zr-MOF-801 MOFs (Fig. 1B), indicating that Ce-MOF-801 MOFs could be easily synthesized in an aqueous solution at various temperatures (25–100 °C). The BET surface areas of all samples were characterized by N_2 sorption tests (Fig. S1, ESI†). As shown in Fig. 1C, Ce-MOF-801 synthesized at 40 °C possesses the highest surface area ($898 \text{ m}^2 \text{ g}^{-1}$), and the surface area of the obtained samples gradually decreases with the enhancement of crystallization temperature from 40 to 100 °C. This result might be attributed to the decrease of material defects caused by the increase of synthesis temperature.¹⁴ Meanwhile, a sharp decrease in the yields (Fig. 1D) of Ce-MOF-801 MOFs occurs with the increase of synthesis temperature from 40 to 100 °C. Therefore, taking the crystallinity, BET surface area, and yields together, 40 °C was optimized to continue the following research.

The influence of reaction time was examined by setting the crystallization time from 10 min to 12 h at 40 °C. The PXRD patterns (Fig. S2, ESI†) show that the structure of Ce-MOF-801 could be rapidly formed within 10 min, and the crystallinity of the samples remains unchanged with the increase of the reaction time. The N_2 sorption isotherms (Fig. S3 and S4, ESI†) demonstrate that a surface area of $763 \text{ m}^2 \text{ g}^{-1}$ can be obtained when the reaction time is only 10 min. With the increase of the reaction time, the maximum surface area ($898 \text{ m}^2 \text{ g}^{-1}$) was achieved at 6 h, indicating the coordination equilibrium between Ce^{4+} ions and the organic ligand of FUM. Since the reaction time has no obvious effect on the yields of the formed MOFs (Fig. S4, ESI†), 6 h was taken as the most suitable reaction time (Table 1).

It is well known that the modulator plays an important role in the crystallization process of MOFs.¹⁵ The effect of the applied amount of the HAc modulator on the crystallinity and porosity of Ce-MOF-801 was also investigated. The N_2 sorption tests show (Fig. S5, ESI†) that the surface area of the samples increases with the increase of the amount of HAc within 70 eq., and the maximum surface area ($898 \text{ m}^2 \text{ g}^{-1}$) is obtained when 70 eq. HAc is applied (Fig. 1D). The effect of HAc on the surface area of samples could be attributed to the increased crystallinity of the Ce-MOF-801 samples, as confirmed from the PXRD pattern in Fig. S6 (ESI†). The slight decrease of the surface area of the sample prepared with 105 eq. HAc could be attributed to the increase of the sample particle size (Fig. S7, ESI†), which results in the decrease of the external surface area (Table S1, ESI†). In addition, the yields of the sample show a significant decrease with the increase of HAc dosage to 100 eq. Therefore, 70 eq. HAc was considered as the optimal value for the synthesis of Ce-MOF-801.

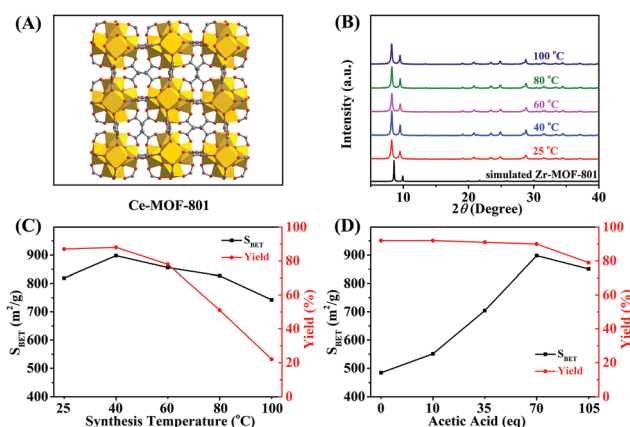


Fig. 1 (A) Illustration of the Ce-MOF-801 structure. PXRD patterns (B), BET surface area and yields (C) of the Ce-MOF-801 synthesized at different temperatures. (D) BET surface area and yields of Ce-MOF-801 synthesized with different amounts of HAc.

Table 1 Summary of the yields of Ce-MOF-801 synthesized under different conditions

Temperature (°C)	Yields (%)	Time	Yields (%)	HAc (eq.)	Yields (%)
25	87	10 min	86	0	92
40	88	30 min	87	10	92
60	78	2 h	86	35	91
80	51	6 h	88	70	90
100	22	12 h	88	105	79

In view of the above results, a reaction temperature of 40 °C, a reaction time of 6 h and 70 eq. HAc are considered to be the best conditions for the synthesis of Ce-MOF-801 with high surface area and good crystallinity. However, to satisfy the increasing need for an energy-saving and environmentally benign procedure for adsorptive separation, a rapid and low temperature separation of RE/Ce is highly desirable. Thanks to the strong crystallization tendency between Ce(IV) and FUM under various conditions, the developed VSC strategy could be implemented at a reaction time of 10 min, a reaction temperature of 25 °C and 70 eq. HAc. To prove this, seven RE ions with the same molar equivalent to Ce ions were applied to the Ce-MOF-801 reaction system to check the separation effect of Ce from RE ion mixtures, and the obtained Ce-MOF-801 samples are denoted as RE/Ce-MOF-801 (RE = La, Pr, Nd, Sm, Tb, Er, and Yb). The PXRD patterns (Fig. S8, ESI†) indicate that the phase purity of the prepared RE/Ce-MOF-801 is not disturbed by other trivalent RE ions. The sorption isotherms (Fig. S9, ESI†) exhibit the retained pore structure of RE/Ce-MOF-801 MOFs. Moreover, compared with RE-free Ce-MOF-801, the surface areas and yields of the RE/Ce-MOF-801 MOFs exhibit almost no remarkable change (Table S2, ESI†), indicating that the crystallization of Ce(IV) and FUM is not affected by other trivalent RE ions. Thus, the structural sensitivity of the pre-designed MOFs to the valence state of the RE ions strongly prompts us to apply them for the separation of RE ions. As shown in Fig. 2, the molar fraction of Ce in RE/Ce-MOF-801 is more than 99.97%, and only traces (molar fraction less than 0.03%) of trivalent RE elements are observed in RE/Ce-MOF-801, which might be attributed to the weak physical adsorption between trivalent RE ions and Ce-MOF-801. This result indicates that Ce⁴⁺ ions can be selectively crystallized into pre-designed MOFs from the RE mixed solution because of their different valence states. Therefore, the high efficiency and excellent specificity for Ce(IV) separation prefigure the great potential of the current approach for the purification of RE elements.

Considering that the realization of amplified separation is of great value for economic production and practical applications, the large-scale separation of Ce and RE elements was demonstrated at

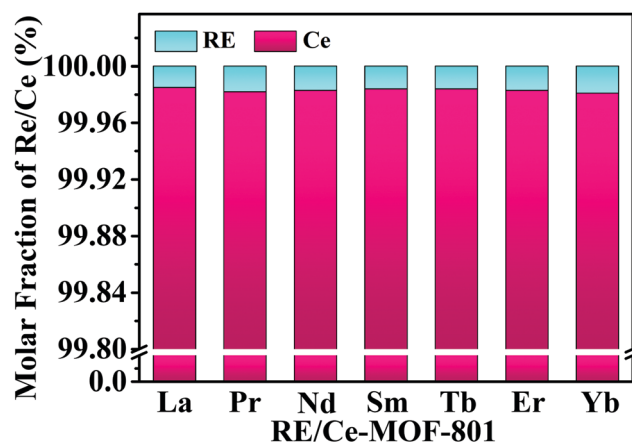


Fig. 2 The molar fraction of RE and Ce in RE/Ce-MOF-801 with a feed ratio of RE : Ce = 1 : 1 (RE = La, Pr, Nd, Sm, Tb, Er and Yb).

room temperature by choosing La³⁺ as the representative RE impurity. The crystallinity of the La/Ce-MOF-801 sample remains unchanged upon enlarging the separation system to 100-fold (Fig. S10, ESI†). The ICP analysis shows that only traces of La (molar fraction less than 0.03%) were found in the obtained 28 g La/Ce-MOF-801 (Fig. S11, ESI†). In addition, the molar fraction of La³⁺ ions in the filtrate accordingly increases from 50% to 91% (Fig. S11, ESI†). Therefore, these results imply that the applied VSC strategy provides a highly efficient separation capability and possesses potential for practical applications in the field of RE separation and purification.

To verify that the selection of organic ligands is not a key factor for the VSC strategy, other green and commercially available organic ligands were also tried, such as SUC, BTC and Na₂BDC. As shown in Fig. 3A, when the three ligands are applied to the La/Ce separation system, the corresponding three designed Ce-MOFs (Ce-Uio-66, Ce-MOF-808, Ce-Uio-66-SUC) are successfully obtained. PXRD patterns (Fig. S12, ESI†) reveal that these Ce-MOFs possess a pure phase structure, indicating that the crystallization of Ce-MOFs is not inhibited in the presence of trivalent La ions. To the best of our knowledge, this is the first report on the aqueous-phase rapid synthesis of such kinds of materials although these Ce-MOFs are traditionally synthesized by the solvothermal method.^{12,16} More importantly, only traces of La (less 0.03%) are detected in

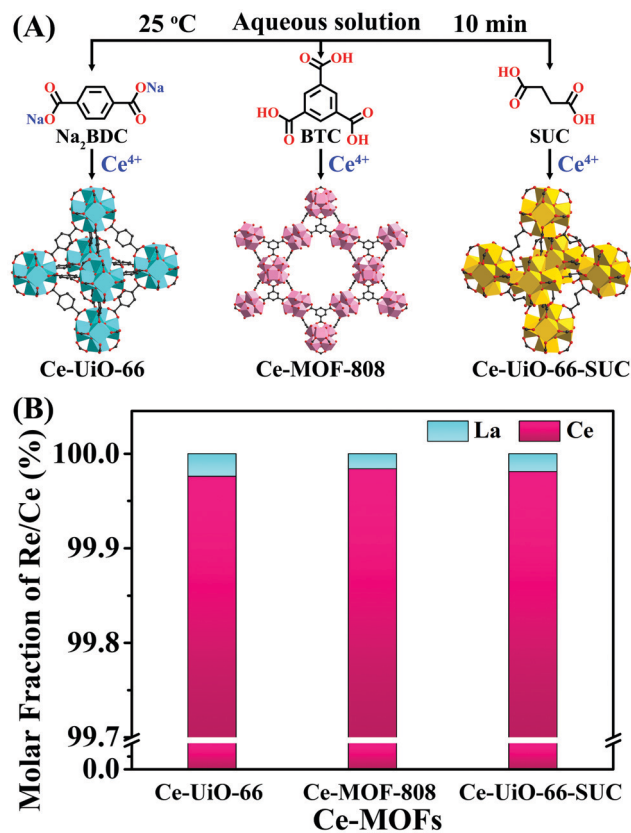


Fig. 3 (A) The aqueous-phase synthesis of three pre-designed Ce-MOFs. (B) The molar fraction of La and Ce in three Ce-MOFs with a feed ratio of La : Ce = 1 : 1.

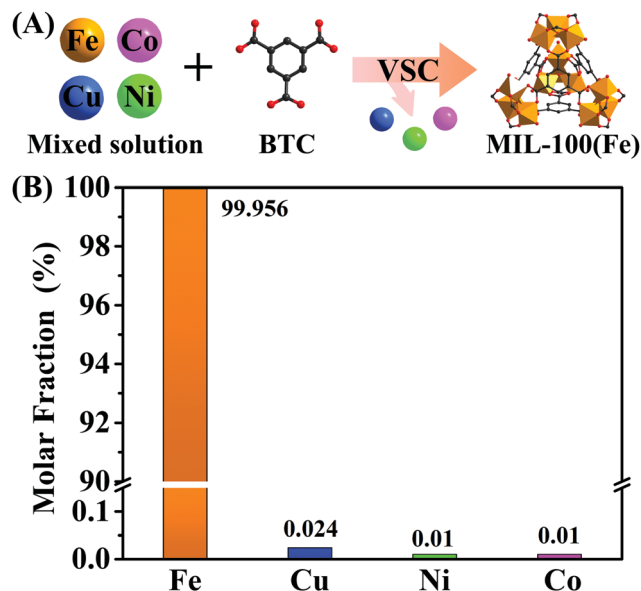


Fig. 4 (A) Precise separation of Fe^{3+} ions is realized from the mixed aqueous solutions of Fe^{3+} , Co^{2+} , Ni^{2+} , and Cu^{2+} ions by the VSC strategy. (B) The molar fractions of Fe, Cu, Ni and Co in the obtained MIL-100(Fe) MOFs.

the obtained La/Ce-MOFs (Fig. 3B), which means that the VSC strategy can be successfully applied by employing diverse ligands as crystallizing agents.

To check the extensibility, the developed VSC strategy was further applied to the selective separation of Fe^{3+} from the mixed solutions of Fe^{3+} , Ni^{2+} , Co^{2+} and Cu^{2+} ions. The commercially available BTC was selected as an organic ligand to crystallize Fe^{3+} into Fe-based MOFs (as demonstrated in Fig. 4A). After the hydrothermal reaction, a brown solid was obtained from the mixed aqueous solution of Fe^{3+} , Ni^{2+} , Co^{2+} and Cu^{2+} ions. PXRD patterns (Fig. S13, ESI[†]) demonstrate that the solid was a pure phase of MIL-100(Fe) MOFs,¹⁷ indicating that the bivalent interference ions engender negligible effects on the crystallization of Fe-MOFs. ICP analysis (Fig. 4B) shows that the molar ratio of Fe in the MIL-100(Fe) MOFs is as high as 99.956%, while the other three divalent ions are only in traces, indicating that the VSC strategy can be successfully extended to the purification and separation of Fe.

In summary, an aqueous-phase synthesis has been explored in detail for the mild preparation of Ce-MOF-801 for the first time. Based on this, a green VSC strategy is developed for the specific separation of Ce(IV) from a RE mixed solution, and the difference of the valence states between Ce(IV) and other RE elements is the key to the success of this approach. The strategy could afford a purification efficiency as high as 99.97% for Ce(IV) in 10 min at room temperature and could remain effective upon enlarging the designed separation system to 100-fold. This strategy also shows a wide selection range of ligands, and several cheap and commercially available ligands can be applied to the RE separation system. Additionally, the VSC strategy is universal to other separation systems so that it could be extended to the separation of Fe from the mixed aqueous solutions of Fe^{3+} , Ni^{2+} , Co^{2+} and Cu^{2+} ions. Therefore, the

currently developed VSC approach is efficient, energy-saving, and environmentally benign, prefiguring a new perspective for application in RE separation.

This work was financially supported by the Natural Science Foundation of China (21975072, 519021065), the Natural Science Foundation of Shanghai (18ZR1408700), Projects of Shanghai Municipality (18JC1410802), the National Key Research and Development Program (2016YFC1102100) and 111 Project (B14018).

Conflicts of interest

There are no conflicts to declare.

Notes and references

- (a) D. Pugliese, N. G. Boetti, J. Lousteau, E. Ceci-Ginistrelli, E. Bertone, F. Geobaldo and D. Milanese, *J. Alloys Compd.*, 2016, **657**, 678–683; (b) H. Dong, S.-R. Du, X.-Y. Zheng, G.-M. Lyu, L.-D. Sun, L.-D. Li, P.-Z. Zhang, C. Zhang and C.-H. Yan, *Chem. Rev.*, 2015, **115**, 10725–10815; (c) G. Wang, P. Qing and Y. Li, *Acc. Chem. Res.*, 2011, **44**, 322–332; (d) R. J. Cava, *Chem. Commun.*, 2005, 5373–5377.
- (a) S. Xie, Z. Wang, F. Cheng, P. Zhang, W. Mai and Y. Tong, *Nano Energy*, 2017, **34**, 313–337; (b) T. Montini, M. Melchionna, M. Monai and P. Fornasiero, *Chem. Rev.*, 2016, **116**, 5987–6041.
- Z. Xia and A. Meijerink, *Chem. Soc. Rev.*, 2017, **46**, 275–299.
- M. Aguirre, M. Paulis and J. R. Leiza, *J. Mater. Chem. A*, 2013, **1**, 3155–3162.
- (a) M. Breitwieser, C. Klose, A. Hartmann, A. Büchler, M. Klingele, S. Vierrath, R. Zengerle and S. Thiele, *Adv. Energy Mater.*, 2017, **7**, 1602100; (b) S. Chaudhary, P. Sharma, D. Singh, A. Umar and R. Kumar, *ACS Sustainable Chem. Eng.*, 2017, **5**, 6803–6816.
- F. Xie, T. Zhang, D. Dreisinger and F. Doyle, *Miner. Eng.*, 2014, **56**, 10–28.
- Y. Inoue, H. Kumagai and Y. Shimomura, *Anal. Chem.*, 1996, **68**, 1517–1520.
- W. C. Wilfong, B. W. Kail, T. L. Bank, B. H. Howard and M. L. Gray, *ACS Appl. Mater. Interfaces*, 2017, **9**, 18283–18294.
- (a) J. Zhang, S. B. Peh, J. Wang, Y. Du, S. Xi, J. Dong, A. Karmakar, Y. Ying, Y. Wang and D. Zhao, *Chem. Commun.*, 2019, **55**, 4727–4730; (b) D. Lin, X. Liu, R. Huang, W. Qi, R. Su and Z. He, *Chem. Commun.*, 2019, **55**, 6775–6778; (c) H.-R. Fu, N. Wang, J.-H. Qin, M.-L. Han, L.-F. Ma and F. Wang, *Chem. Commun.*, 2018, **54**, 11645–11648; (d) Y. Bai, Y. Dou, L.-H. Xie, W. Rutledge, J.-R. Li and H.-C. Zhou, *Chem. Soc. Rev.*, 2016, **45**, 2327–2367.
- (a) P. Yang, M. Gong, Y. Ye, Y. Li, Q. Zhuang and J. Gu, *Chem. – Asian J.*, 2019, **14**, 135–140; (b) Y.-Q. Hu, T. Zhang, M.-Q. Li, Y. Wang, Z. Zheng and Y.-Z. Zheng, *Green Chem.*, 2017, **19**, 1250–1254; (c) H. Gao, W. Peng, P. Meng, X. Feng, J. Li, H. Wu, C. Yan, Y. Xiong and F. Luo, *Chem. Commun.*, 2017, **53**, 5737–5739; (d) X. Zhao, M. Wong, C. Mao, T. X. Trieu, J. Zhang, P. Peng and X. Bu, *J. Am. Chem. Soc.*, 2014, **136**, 12572–12575.
- F. Jiang, S. Yin, C. Srinivasakannan, S. Li and J. Peng, *Chem. Eng. J.*, 2018, **334**, 2208–2214.
- M. Lammert, M. T. Wharmby, S. Smolders, B. Bueken, A. Lieb, K. A. Lomachenko, D. De Vos and N. Stock, *Chem. Commun.*, 2015, **51**, 12578–12581.
- G. C. Shearer, S. Chavan, J. Ethiraj, J. G. Vitillo, S. Svelle, U. Olsbye, C. Lamberti, S. Bordiga and K. P. Lillerud, *Chem. Mater.*, 2014, **26**, 4068–4071.
- (a) M. R. DeStefano, T. Islamoglu, S. J. Garibay, J. T. Hupp and O. K. Farha, *Chem. Mater.*, 2017, **29**, 1357–1361; (b) A. M. Ploskonka, S. E. Marzen and J. B. DeCoste, *Ind. Eng. Chem. Res.*, 2017, **56**, 1478–1484.
- (a) W. Morris, S. Wang, D. Cho, E. Auyeung, P. Li, O. K. Farha and C. A. Mirkin, *ACS Appl. Mater. Interfaces*, 2017, **9**, 33413–33418; (b) A. Schaate, P. Roy, A. Godt, J. Lippeke, F. Waltz, M. Wiebecke and P. Behrens, *Chem. – Eur. J.*, 2011, **17**, 6643–6651.
- M. Lammert, C. Glißmann, H. Reinsch and N. Stock, *Cryst. Growth Des.*, 2017, **17**, 1125–1131.
- M. Tong, D. Liu, Q. Yang, S. Devautour-Vinot, G. Maurin and C. Zhong, *J. Mater. Chem. A*, 2013, **1**, 8534–8537.